authorizationToken

Service Description

Service ID: *"authorizationToken"*

Abstract
This document describes the `authorizationToken` service, version 1.0.0, produced by the ConsumerAuthorization core system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface **4**
3.1 operation generate . . . 4
3.2 operation verify . . . 4
3.3 operation get-public-key . . . 4
3.4 operation register-encryption-key . . . 4
3.5 operation unregister-encryption-key . . . 4

4 Information Model **5**
4.1 struct AuthorizationEncryptionKeyRegistrationRequest . . . 5
4.2 struct AuthorizationTokenGenerationRequest . . . 5
4.3 struct AuthorizationTokenGenerationResponse . . . 5
4.4 struct AuthorizationTokenVerifyResponse . . . 5
4.5 Primitives . . . 6

5 References **7**

6 Revision History **8**
6.1 Amendments . . . 8
6.2 Quality Assurance . . . 8

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `authorizationToken` service, which provides authorization tokens: a consumer obtains a token for a specific provider, service and operation; the provider verifies it. It is produced by the ConsumerAuthorization system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Successor of the Arrowhead 4.x token generation of the Authorization system (which issued JWTs during orchestration).

2.2 How This Service Is Meant to Be Used

Normally DynamicServiceOrchestration requests the token on behalf of the consumer during orchestration, when the provider interface has a token policy. A consumer may also call `generate` itself. The provider calls `verify` (simple tokens) or validates a JSON Web Token locally with the key from `get-public-key`. Providers that want encrypted self-contained tokens register an AES key with `register-encryption-key`.

2.3 Important Delimitations and Dependencies

A token is issued only if the consumer is authorized by the policies. Simple tokens are limited by time (`token.time.limit`) or number of uses (`simple.token.usage.limit`). The service does not transport tokens; the consumer sends them in its requests to the provider.

3 Service Interface

This section describes the interfaces to the `authorizationToken` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `authorizationToken` service and its operations.

The following interface operations are available.

3.1 operation generate (AuthorizationTokenGenerationRequest) : AuthorizationTokenGenerationResponse

Generates a token of the requested variant for the requester (as consumer) to the provider's target and operation, if authorized.

3.2 operation verify (Token) : AuthorizationTokenVerifyResponse

Verifies a token presented to the requester (as provider), counts a use of usage-limited tokens and returns its consumer, target and scope.

3.3 operation get-public-key () : String

Returns the public key (Base64 X.509 SubjectPublicKeyInfo) with which providers verify JSON Web Tokens.

3.4 operation register-encryption-key (AuthorizationEncryptionKeyRegistrationRequest) : String

Registers or replaces the requester's AES key for encrypting its self-contained tokens. Returns the initialization vector (CBC) or an empty string; 201 for a new key.

3.5 operation unregister-encryption-key ()

Removes the requester's encryption key. Returns 200 if removed, 204 if there was none.

4 Information Model

Here, all data objects that can be part of the `authorizationToken` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.5, which are used to represent things like hashes and identifiers.

4.1 struct AuthorizationEncryptionKeyRegistrationRequest

Registers the requester's key for encrypting its self-contained tokens.

Field	Type	Description
key	String	AES key of the provider (at least 16 bytes).
algorithm	String	Encryption algorithm of the key, e.g. `AES/ECB/PKCS5Padding` or `AES/CBC/PKCS5Padding`.

4.2 struct AuthorizationTokenGenerationRequest

Request of the consumer for a token to one provider's service operation.

Field	Type	Description
tokenVariant	String	Requested token variant, as interface security policy (e.g. `TIME_LIMITED_TOKEN_AUTH`, `RSA_SHA512_JSON_WEB_TOKEN_AUTH`).
provider	Name	Provider system the token is for.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Service definition (or event type).
scope	Name	Operation the token is for.

4.3 struct AuthorizationTokenGenerationResponse

A generated token.

Field	Type	Description
tokenType	AuthorizationTokenType	Token type: `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN` or `TRANSLATION_BRIDGE_TOKEN`.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
token	Token	The token (opaque string, Base64 or JSON Web Token).
usageLimit	Number	Number of allowed uses (usage-limited tokens).
expiresAt	DateTime	UTC expiry of time-limited tokens.

4.4 struct AuthorizationTokenVerifyResponse

Result of verifying a token presented to a provider.

Field	Type	Description
verified	Boolean	True when the token is valid.
consumerCloud	Name	Cloud identifier of the consumer system.
consumer	Name	Name of the consumer system.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
scope	Name	Operation the token or policy is limited to; empty for all operations.

4.5 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
AuthorizationTargetType	One of `SERVICE_DEF`, `EVENT_TYPE`.
AuthorizationTokenType	One of `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN`, `TRANSLATION_BRIDGE_TOKEN`.
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.
Token	An opaque string used as credential.

5 References

[1] J. Delsing, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, consumerauthorization/documentation/v5.3.0/ConsumerAuthorization-v5.3.0-SysD.pdf.

[2] ——, “authorizationToken HTTP/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationToken-HTTP-IDD.pdf.

[3] ——, “authorizationToken MQTT/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationToken-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	